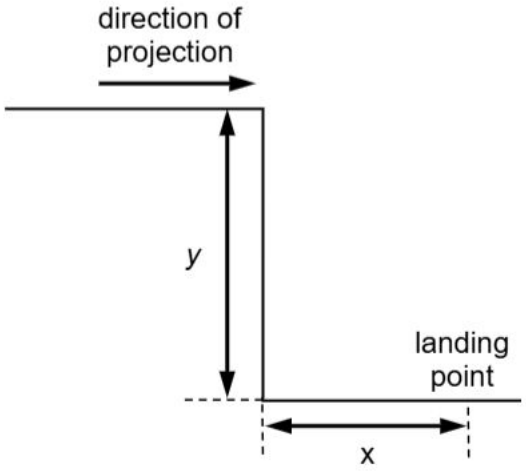
8 An object is projected horizontally. The object falls a vertical distance y and travels a horizontal distance x before landing on the ground.



A second object is projected horizontally with the same initial velocity and falls a vertical distance $4y$ before landing on the ground.

Assume that air resistance is negligible.

Which horizontal distance does the second object travel?

A x **B** $2x$ **C** $4x$ **D** $16x$

Ans: **B**

First object,

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$y = \frac{1}{2} g t_1^2$$

$$t_1 = \sqrt{\frac{2y}{g}}$$

Second object,

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$4y = \frac{1}{2} g t_2^2$$

$$t_2 = \sqrt{\frac{2(4y)}{g}} = 2\sqrt{\frac{2y}{g}} = 2t_1$$

With constant horizontal velocity, twice the time of travel means twice the horizontal distance, 2*x*.

9 Body X of mass 2.5 kg undergoes a head-on collision with body Y of mass 1.0 kg. Bodies X and